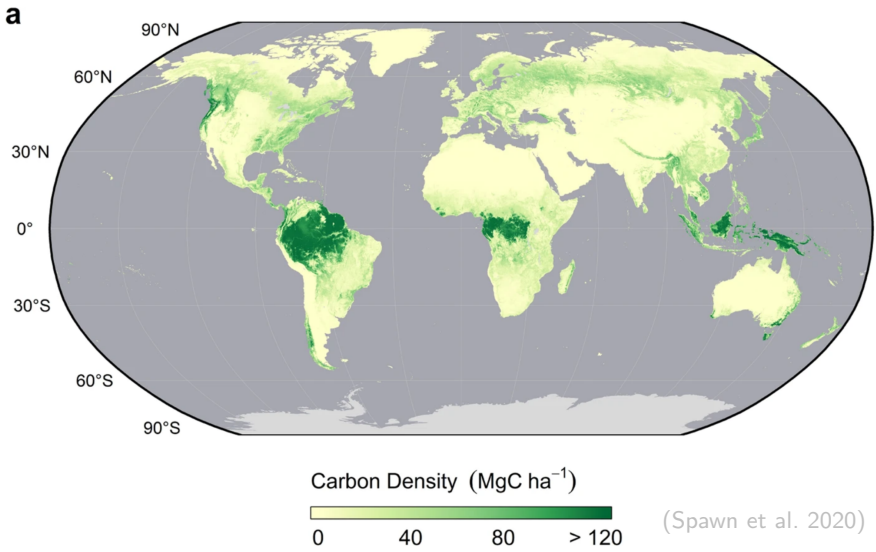
Conservation Policy with Political Resistance

Robin Burgess, LSE
Francisco Costa, FGV EPGE
Allan Hsiao, Stanford
Benjamin Olken, MIT
Veronica Salazar Restrepo, IIES

November 4, 2026

Forest conservation protects major carbon stocks



(Spawn et al. 2020)

But conservation is political

New Internationalist

View from Brazil: Agribusiness lobby scuppers climate gains

Lula wants Brazil to be a beacon in the fight against global warming, Leonardo Sakamoto. The powerful lobby that represents agribusiness in...

Sep 4, 2023

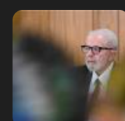


Mother Jones

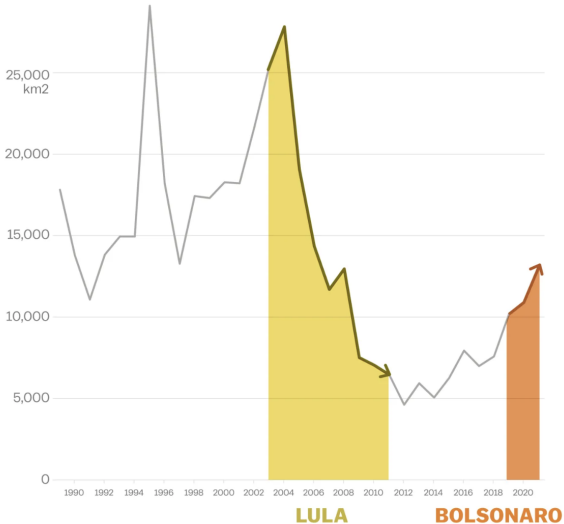
Brazil's Lula Made Progress on Deforestation, but "Agribusiness Is Winning"

When Brazilian president Luiz Inácio Lula da Silva took office in January 2023, he inherited environmental protection agencies in shambles and deforestation at...

Nov 29, 2024



And politics threatens conservation





Deforestation under Bolsonaro



Total area of Belgium

(Northrop 2022)

Question

How can environmental regulation navigate political resistance?

This paper

1. Regulation faces resistance
   - In Brazil, conservation mobilized agricultural interests
2. Resistance constrains policy
   - Targeting where abatement is high and burden is low
3. Optimal policy navigates resistance
   - Balancing welfare gains against repeal risk

Regulation faces resistance

The Brazilian Amazon

- **PPCDAm** strengthened forest protections beginning in 2004
  - Forest Code requires 80% of each property to remain forested
  - Satellite monitoring, protected areas, and tougher penalties
- Deforestation fell sharply through 2012
  - And then rose again as enforcement weakened

Measuring political resistance

	Donors	Candidates
Full sample	6,967	3,809
Agricultural firm owners	765	107
Environmental infractors	306	725

- State and federal legislative elections from 2002 to 2018
  - Universe of formal campaign donations (TSE)
  - Linked to firm ownership (*Receita Federal*)
  - And to environmental infractions (IBAMA)

Agricultural candidates supported weaker forest policy

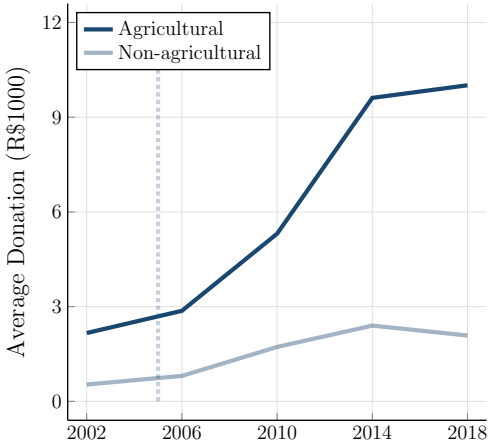
	Forest Code roll calls		
	Reform May 2011	Amendment May 2011	Provisions April 2012
Ag	3.11	9.04**	11.22***
	(2.18)	(4.00)	(3.97)
Effect as % of mean	3	11	13
Observations	471	454	453
Fixed effects	$s + p$	$s + p$	$s + p$

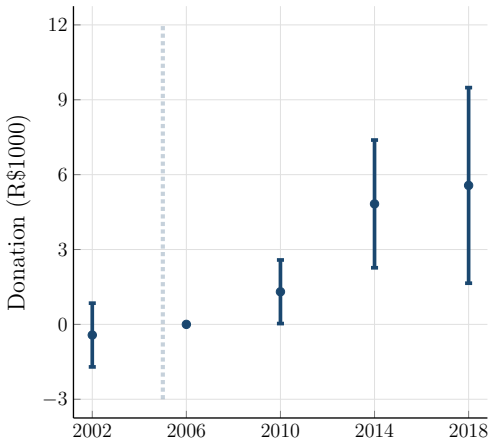
Difference-in-differences

$$y_{ist} = \sum_{\tau \neq 2006} \beta_\tau \, \text{Agricultural}_i \times \mathbb{1}\{t = \tau\} + \alpha_i + \gamma_{st} + \varepsilon_{ist}$$

- Agricultural vs. non-agricultural, before vs. after PPCDAm
- Individuals i, states s, election years t
  - Individual and state-by-year fixed effects
  - Matching on firm age, structure, size, capital, tax regime (entropy balancing)

PPCDAm mobilized agricultural donations





Agricultural donors gave more

	All candidates	Agricultural candidates
$Ag \times Post$ 2006	3.02***	2.07***
	(0.95)	(0.55)
Effect as % of mean	120	180
Observations	34,820	34,820
Fixed effects	$i + st$	$i + st$

Especially where enforcement was binding

	Legal Amazon	Cloud coverage
$Ag \times Post$ 2006	0.69**	1.75*
	(0.32)	(0.90)
$Ag \times Post$ 2006 $\times H$	2.33**	-3.08***
	(1.00)	(0.98)
H effect as % of mean	93	-121
Observations	372,055	27,785
Fixed effects	$i + st$	$i + st$

Agricultural candidates gained in state elections

	Elected		Votes	
	Federal	State	Federal	State
$Ag \times Post$ 2006	−0.61	24.14***	−38.04	22.10**
	(9.07)	(8.65)	(25.50)	(10.64)
Effect as % of mean	−1	61	−29	25
Observations	681	2,418	528	2,035
Fixed effects	$i + st$	$i + st$	$i + st$	$i + st$

Again where enforcement was binding

	Elected		Votes	
	Federal	State	Federal	State
$Ag \times Post$ 2006	−7.61	0.32	−26.48	−14.86
	(4.71)	(4.86)	(23.59)	(9.29)
$Ag \times Post$ 2006 $\times Amazon$	6.89	23.78**	−11.57	36.81***
	(10.08)	(9.90)	(34.28)	(14.10)
Amazon effect as % of mean	9	60	−9	42
Observations	4,133	9,017	3,164	7,066
Fixed effects	$i + st$	$i + st$	$i + st$	$i + st$

Resistance constrains policy

The Pigouvian benchmark

$$L = \{i \mid v_i < \kappa e_i\}, \qquad H = \{i \mid v_i > \kappa e_i\}$$

- Each plot i produces or conserves
  - Production generates agricultural value $v_i > 0$
  - Clearing releases emissions e_i at social cost κ
- Pigouvian tax κe_i maximizes welfare
  - Efficient H produces, inefficient L does not

Abatement and burden can diverge

$$L = \{i \mid v_i < \kappa e_i\}, \qquad H = \{i \mid v_i > \kappa e_i\}$$

- Pigouvian tax κe_i
  - L stops producing: abatement is e_i, burden is v_i
  - H keeps producing: abatement is 0, burden is κe_i

- Targeted tax κe_i on L, exempting H
  - Same production and abatement
  - While eliminating burden on H

Quantifying policy impacts

- We build a simple empirical model
  - Characterizes value (targeting)
  - Predicts land use responses (abatement)
  - Predicts producer surplus losses (burden)

- Do higher agricultural stakes predict larger donation responses?
- How effective are policies that account for agricultural interests?

Agricultural land use reveals value

$$\max\left\{v_i - \tau_i + \epsilon^a_{ij},\, \epsilon^f_{ij}\right\}, \qquad \log\left(\frac{a_i}{f_i}\right) = \frac{v_i - \tau_i}{\sigma}$$

- Site i allocates each hectare j to agriculture or forest
  - Adding logit shocks with scale σ (continuous shares)
- Land use shares reveal value relative to σ

Agricultural returns identify scale

$$\log\left(\frac{a_{it}}{f_{it}}\right) = \beta \underbrace{y_i p_t}_{\text{returns}} + \alpha_i + \gamma_t + \varepsilon_{it}, \qquad \sigma = \frac{1}{\beta}$$

- Coefficient β sets the monetary scale
  - Governs land use responses to returns (and taxes)
  - Gives producer surplus in dollars

- **Identification:** local crop yields $\times$ global price changes
  - Pooled 10-year long differences capture gradual adjustment
  - Spatial panel of 25-km cells in Legal Amazon from 2001 to 2019

Agricultural land use responds to returns

	Crops: soy, maize, pasture		
	Baseline shares	Leave-out shares	Fitted shares
Agricultural returns	0.10**	0.11**	0.10**
	(0.04)	(0.05)	(0.05)
Observations	72,636	72,636	72,636
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$

Abatement and burden in the data

$$S'_m = z\big(\log(S_m)\big), \qquad S_m = \frac{L_m}{N_m}\log\left(1 + \frac{A_m}{F_m}\right)$$

- Abatement: carbon density per hectare (observed)
- Burden: implied surplus per producer (donor)
  - Logit measure with pre-period municipal data
  - Land use reveals agricultural value

Resistance rises with burden, not abatement

	Implied surplus	Carbon density	Both
$Ag \times Post$ 2006	1.92**	3.11***	1.90**
	(0.86)	(0.96)	(0.90)
$Ag \times Post$ 2006 $\times S$	3.01***		3.03**
	(1.05)		(1.19)
$Ag \times Post$ 2006 $\times E$		−1.12	0.05
		(1.14)	(1.27)
S effect as % of mean	119		120
E effect as % of mean		−44	2
Observations	34,555	34,555	34,555
Fixed effects	$i + st$	$i + st$	$i + st$

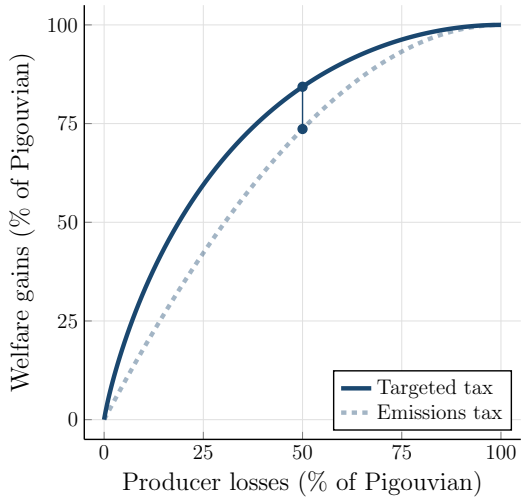
Policy with political constraints

$$\mathcal{W}(\bar{B}) = \max_{\{\tau_i\}} W(\tau) \quad \text{s.t.} \quad B(\tau) \leq \bar{B}$$

- Burden limit $\bar{B}$ traces out the welfare frontier
- Want to target high emissions e_i, low burden v_i

Targeting expands the welfare frontier

- Welfare frontier $\mathcal{W}(\bar{B})$
  - Varying burden limit $\bar{B}$
  - Optimize taxes across locations
- Common emissions tax
  - Same tax per ton everywhere
  - Lower welfare for same burden
- At half the Pigouvian burden
  - Targeted welfare gains +15%



Optimal policy navigates resistance

Resistance can undo regulation

$$F(\tau) = \underbrace{[1 - R(B(\tau))]\, W(\tau)}_{\text{regulation}} + \underbrace{R(B(\tau))\, W(0)}_{\text{repeal}}$$

- Regulation survives or is repealed
  - Regulation imposes total burden $B(\tau)$
  - Burden raises repeal risk with $R'(B) > 0$

- Optimal policy maximizes **feasible welfare** $F(\tau)$
  - Balances welfare gains against repeal risk

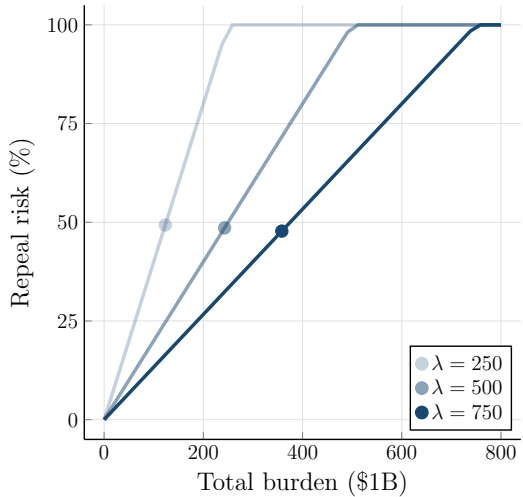
Resistance narrows the Pigouvian tax

$$\tau_i^* = \underbrace{\kappa e_i}_{\text{Pigouvian}} \cdot \mathbb{1}\left[\frac{v_i}{\kappa e_i} < \frac{1}{1+\mu^*}\right]$$

- Pigouvian with a political cutoff
  - Regulate only where emissions are high relative to value
  - Exempt all efficient H and some inefficient L
- Resistance tightens the cutoff
  - Added burden has shadow cost $\mu^* > 0$

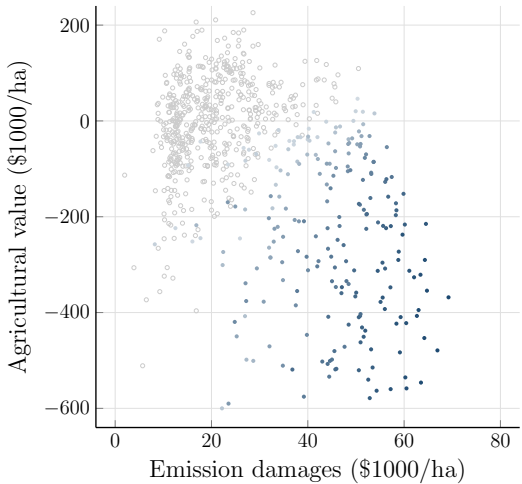
Optimal policy depends on repeal risk

- The land use model gives $B(\tau)$
  - Burden raises donations
- We specify repeal risk $R(B)$
  - Linear (locally) up to λ
- Dots maximize feasible welfare
  - Higher λ allows greater burden



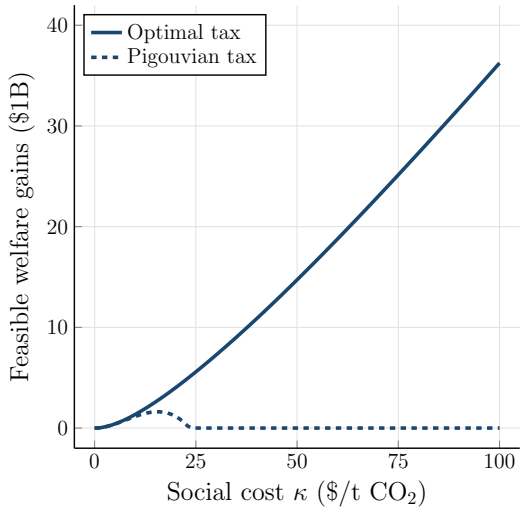
Optimal policy targets both emissions and value

- Pigouvian with political shading
  - Pigouvian helps welfare gains
  - Shading helps repeal risk
- Avoids low-e, high-v munis (lighter)
  - Low welfare gains
  - High repeal risk
- Targets high-e, low-v munis (darker)



Optimal policy avoids repeal

- Pigouvian tax collapses
  - High burden $\rightarrow$ certain repeal
- Optimal tax survives
  - With targeted shading
  - Welfare gains grow in κ

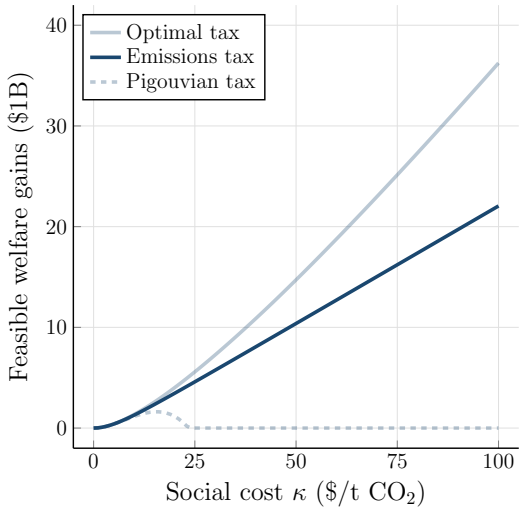


Can practical policies realize the gains from targeting?

- Optimal policy targets emissions and value
- Practical policies use less information
  - Emissions are observable via satellites
  - But value is private information

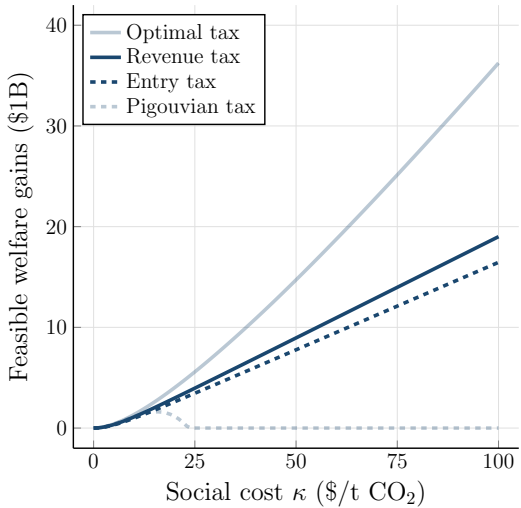
Practical policies capture substantial gains

- Emissions taxes are best
  - Targets e with $\text{corr}(e, v) = -0.35$
  - Reduced tax rate for large κ
- Revenue/entry taxes are good
  - But do not directly target e
- Emissions bans fail, even on the PPF
  - No revenue + producer losses $> \kappa e$



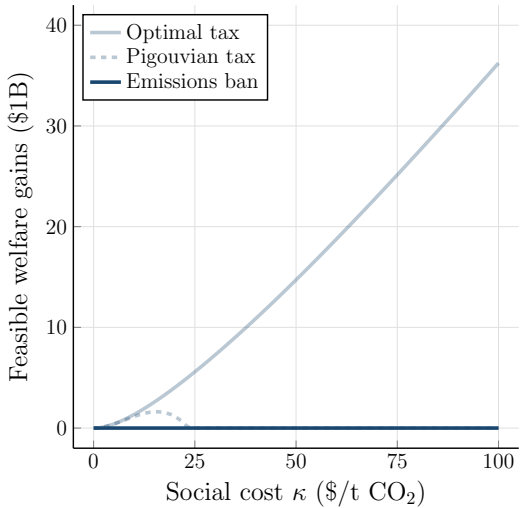
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Conclusion

Conservation is political

- In Brazil, conservation sparked resistance
  - Agribusiness responded with money and votes
- Optimal policy is politically targeted